

EEE 416 (January 2025)
Microprocessor and Embedded System Laboratory

Final Project Report

Section: A2 Group: 06

Electronic Voting Machine with Fingerprint
Authentication

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1 Abstract

In an era where elections form the backbone of democratic processes, the integrity, accuracy, and transparency of voting systems have become paramount. Traditional voting methods are often prone to errors, fraud, and manipulation, which compromise the fairness and reliability of election outcomes. To address these issues, we designed and implemented a fingerprint-authenticated electronic voting machine (EVM), which leverages Arduino-based microcontrollers to create a user-friendly and secure voting experience.

The system's primary goal is to improve voting accuracy by eliminating manual errors and providing tamper-resistant voting using biometric verification. Through serial communication, the selected vote is sent from the slave device to the master, ensuring a seamless and efficient voting process. This project offers a scalable solution that can be easily replicated for future elections in institutional or community settings, emphasizing transparency, security, and accessibility. Environmentally, the project advocates for sustainability by using low-cost, reusable components that reduce the reliance on paper ballots and other materials, aligning with eco-friendly practices.

2 Introduction

In many countries, traditional voting systems still rely heavily on manual processes such as paper ballots and voter lists. While these methods have been used for years, they come with several serious problems. First, manual voting can lead to human errors, such as counting mistakes, mis-recorded votes, or misplaced ballots. Second, these systems are vulnerable to voter impersonation and fraud, where one person may try to vote more than once or vote using someone else's identity. Third, the process is often slow, especially when counting votes, which can delay election results. Additionally, printing ballots and using paper increases costs and environmental waste.

To solve these issues, our project presents a modern, electronic voting machine (EVM) that uses fingerprint authentication for secure voter identification and digital candidate selection using a rotary encoder and TFT display. This makes the voting process faster, safer, and more reliable.

Our system is divided into two parts: a Slave Device and a Master Device. The Slave Arduino is responsible for:

- Verifying the voter's identity using a fingerprint sensor
- Displaying the list of candidates on a TFT screen
- Letting the voter choose a candidate using a rotary encoder
- Sending the selected vote to the Master device via UART serial communication

The Master Arduino receives the vote and:

- Updates the vote count
- Displays live results on another TFT screen

This system prevents unauthorized voting, ensures that each person votes only once, and allows for real-time vote counting. The use of a fingerprint sensor greatly increases the security and fairness of the election. The rotary encoder and touchscreen interface make the system easy to use, even for people unfamiliar with digital devices.

Our project was built using Arduino programming language which is a simplified version of C++, and hardware components like Arduino boards, fingerprint sensor, TFT display, rotary encoder, and serial communication protocols. We made sure to follow all safety and ethical standards, and we designed the system to be cost-effective, environmentally friendly, and easy to replicate in schools, institutions, or small community elections.

3 Design

3.1 Problem Formulation (PO(b))

The democratic process of elections is often challenged by manual errors, impersonation, and time-consuming vote counting. In traditional voting systems, physical ballots and manual authentication leave room for fraud, inefficiency, and logistical overhead. To address these issues, we proposed an Electronic Voting Machine (EVM) with Biometric Authentication, integrating fingerprint recognition and digital communication to ensure secure, fast, and transparent voting.

Our goal was to design and implement a system that eliminates impersonation, prevents duplicate votes, ensures real-time vote tracking, and provides a user-friendly interface. The system architecture separates voter interaction (Slave Device) from vote storage and display (Master Device), connected via UART communication. Each voter's identity is verified using a fingerprint sensor before allowing access to the voting interface, ensuring one person, one vote.

This system not only enhances transparency and security but also supports environmental sustainability by eliminating paper ballots. With real-time vote updates and automated counting, it provides rapid and reliable results, making it highly suitable for institutional elections or small-scale government voting setups.

3.2 Circuit Diagram

The project is built using a Master-Slave architecture, where:

Slave Device handles:

- Fingerprint authentication
- Candidate selection via a rotary encoder
- Vote confirmation
- Encoding and sending vote data via UART

Master Device handles:

- Receiving and decoding vote data
- Live vote counting
- Displaying real-time results

Block Diagram:

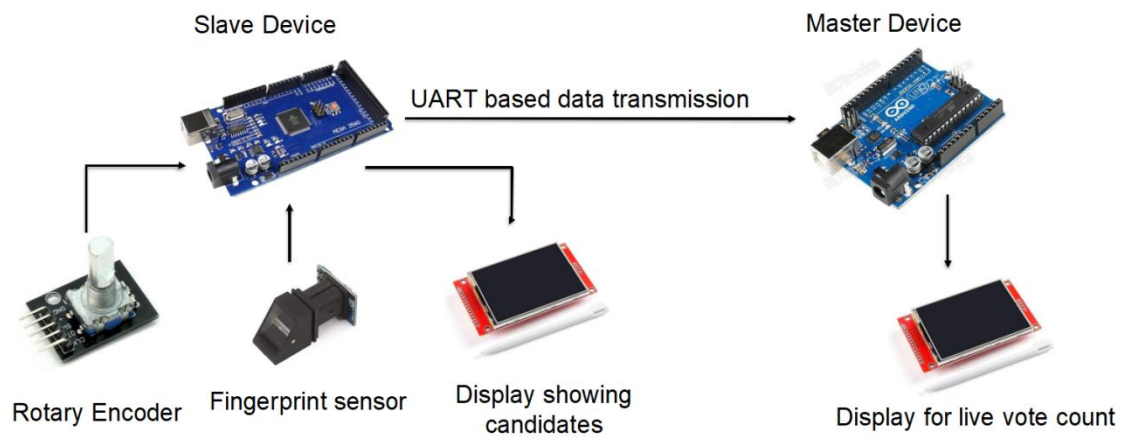


Figure: Block Diagram of the entire project

Each device is implemented using microcontrollers, and communication is achieved using serial transmission (UART). The fingerprint module authenticates users, and only upon successful authentication is the voting interface enabled.

3.3 Full Source Code of Firmware

<pre>//Slave Arduino Code #include "SPI.h" #include <Adafruit_GFX.h> #include <Adafruit_ILI9341.h> #include <Adafruit_Fingerprint.h> #if (defined(__AVR__) defined(ESP8266)) && !defined(__AVR_ATmega2560__) #else // On Leonardo/M0/etc, others with hardware serial, use hardware serial! // #0 is green wire, #1 is white #define mySerial Serial3 #endif //SoftwareSerial softSerial(4, 3); // RX, TX (only TX used) Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial); // === SPI Pin Definitions for Arduino Mega === #define TFT_CS A5 #define TFT_RST A4 #define TFT_DC A3 #define TFT_MOSI 48 #define TFT_CLK 50 #define TFT_MISO 52 int encoderPinA = 5; // CLK pin int encoderPinB = 6; // DT pin int encoderBtn = 7; // SW pin int count = 0; const int voters_count=9; int encoderPinA_prev; int encoderPinA_value; int voters[voters_count]={0}; int current_voter=-1; int matched=0; boolean bool_CW; // Use full 6-argument constructor for Mega Adafruit_ILI9341 tft = Adafruit_ILI9341(TFT_CS, TFT_DC, TFT_MOSI, TFT_CLK, TFT_RST, TFT_MISO); // === Candidate list === const char* candidates[] = { "Alice Ahmed", "Shibli Noman", "Chayan Chowdhury", "Dalia Khatun" }; const unsigned char coffee_logo [] PROGMEM = {...}; //64x64 pixel values const unsigned char shapla_logo [] PROGMEM = {...}; //64x64 pixel values const unsigned char elephant_logo [] PROGMEM = {...}; //64x64 pixel values const unsigned char bird_logo [] PROGMEM = {...}; //64x64 pixel values const int totalCandidates = sizeof(candidates) / sizeof(candidates[0]); int selectedCandidate = 0;</pre>	<pre>Serial.println(candidates[selectedCandidate]); displayCandidates(); delay(1000); encoderPinA_value = digitalRead(encoderPinA); } //displayCandidates(); if (digitalRead(encoderBtn) == LOW) { Serial.print(candidates[selectedCandidate]); Serial.println(" is selected!"); Serial.println(" "); Serial.write(selectedCandidate+1); delay(5000); selectedCandidate=0; displayCandidates(); delay(2000); encoderPinA_value = digitalRead(encoderPinA); matched=0; voters[current_voter]=1; } encoderPinA_prev = encoderPinA_value; } } void displayCandidates() { tft.fillScreen(ILI9341_BLACK); // Clear screen if(selectedCandidate==0)tft.drawBitmap(10, 30, coffee_logo,64,64,ILI9341_WHITE); else if(selectedCandidate==1)tft.drawBitmap(10, 30, shapla_logo,64,64,ILI9341_WHITE); else if(selectedCandidate==2)tft.drawBitmap(10, 30, elephant_logo,64,64,ILI9341_WHITE); else if(selectedCandidate==3)tft.drawBitmap(10, 30, bird_logo,64,64,ILI9341_WHITE); tft.setCursor(10, 140); tft.setTextColor(ILI9341_WHITE, ILI9341_BLACK); tft.println(candidates[selectedCandidate]); } uint8_t getFingerprintID(){ uint8_t p = finger.getImage(); if (p != FINGERPRINT_OK) return 0; p = finger.image2Tz(); if (p != FINGERPRINT_OK) return 0; p = finger.fingerFastSearch(); if (p != FINGERPRINT_OK) return 0; return finger.fingerID; }</pre>
---	--

```

void setup() {
  Serial.begin(9600);
  Serial1.begin(9600);
  //softSerial.begin(9600);
  pinMode (encoderPinA, INPUT);
  pinMode (encoderPinB, INPUT);
  pinMode(encoderBtn, INPUT_PULLUP);

  encoderPinA_prev = digitalRead(encoderPinA);
  Serial.println("Initializing display...");

  tft.begin();
  tft.setRotation(0); // Landscape
  tft.fillScreen(ILI9341_BLACK);
  tft.setTextWrap(false);
  tft.setTextSize(2);

  Serial.println("Display ready.");
  displayCandidates();

  finger.begin(57600);
  if (finger.verifyPassword()) {
    Serial.println("Found fingerprint sensor!");
  } else {
    Serial.println("Did not find fingerprint sensor :(");
    while (1) { delay(1); }
  }
}

void loop() {
  //delay(2000); // Simulate changing selection every 2s

  int id=getFingerprintID();
  if(id!=0 && voters[id]==0){
    matched=1;
    current_voter=id;
    tft.fillScreen(ILI9341_BLACK);
    tft.setCursor(10, 30);
    tft.setTextColor(ILI9341_WHITE, ILI9341_BLACK);
    tft.println("You can vote");
    Serial.println("You can vote! ");
    Serial.println(" ");
    delay(4000);
    displayCandidates();
  }else if(voters[id]!=0){
    tft.fillScreen(ILI9341_BLACK);
    tft.setCursor(10, 30);
    tft.setTextColor(ILI9341_WHITE, ILI9341_BLACK);
    tft.println("Already Voted!!!!");
    Serial.println("Already Voted!!!!");
    Serial.println(" ");
    delay(4000);
    displayCandidates();
  }
  if(matched==1){
    encoderPinA_value = digitalRead(encoderPinA);
    if (encoderPinA_value != encoderPinA_prev) { // check if knob is rotating
      // if pin A state changed before pin B, rotation is clockwise
      if (digitalRead(encoderPinB) != encoderPinA_value) {
        selectedCandidate++;
        if(selectedCandidate>=totalCandidates)selectedCandidate=0;
      } else {
        // if pin B state changed before pin A, rotation is counter-clockwise
        selectedCandidate--;
        if(selectedCandidate<0)selectedCandidate=totalCandidates-1;
      }
      // if(selectedCandidate>=totalCandidates)selectedCandidate=0;
      // if(selectedCandidate<=0)selectedCandidate=totalCandidates-1;
    }
  }
}

```

<pre>//Master Arduino Code #include "SPI.h" #include <Adafruit_GFX.h> #include <Adafruit_ILI9341.h> #include <SoftwareSerial.h> #define TFT_CS A5 #define TFT_RST A4 #define TFT_DC A3 #define TFT_MOSI 11 #define TFT_CLK 12 #define TFT_MISO 13 Adafruit_ILI9341 tft = Adafruit_ILI9341(TFT_CS, TFT_DC, TFT_MOSI, TFT_CLK, TFT_RST, TFT_MISO); const char* candidates[] = { "Alice Ahmed", "Shibli Noman", "Chayan Chowdhury", "Dalia Khatun" }; int selectedCandidate=0; const unsigned char coffee_logo [] PROGMEM = {...}; //64x64 pixel values const unsigned char shapla_logo [] PROGMEM = {...}; //64x64 pixel values const unsigned char elephant_logo [] PROGMEM = {...}; //64x64 pixel values const unsigned char bird_logo [] PROGMEM = {...}; //64x64 pixel values const int totalCandidates = sizeof(candidates) / sizeof(candidates[0]); int votes[totalCandidates]; SoftwareSerial softSerial(4, 3); // RX, TX (only TX used) void setup() { // put your setup code here, to run once: Serial.begin(9600); softSerial.begin(9600); for(int i=0;i<totalCandidates;i++)votes[i]=0; Serial.println("Initializing display..."); tft.begin(); tft.setRotation(0); // Landscape tft.fillScreen(ILI9341_BLACK); tft.setTextWrap(false); tft.setTextSize(2); Serial.println("Display ready."); displayCandidates(); } void loop() { // put your main code here, to run repeatedly: if(softSerial.available()){ int r=softSerial.read(); Serial.print("Vote for "); Serial.println(candidates[r-1]); votes[r-1]++; Serial.println("Total Votes: "); for(int i=0;i<totalCandidates;i++){ Serial.print(candidates[i]); Serial.print(": "); Serial.print(votes[i]); Serial.println(" votes"); } } Serial.println(" "); } selectedCandidate++; if(selectedCandidate==totalCandidates) selectedCandidate=0; displayCandidates(); delay(5000); } void displayCandidates() { tft.fillScreen(ILI9341_BLACK); // Clear screen if(selectedCandidate==0)tft.drawBitmap(10, 30, coffee_logo,64,64,ILI9341_WHITE); else if(selectedCandidate==1)tft.drawBitmap(10, 30, shapla_logo,64,64,ILI9341_WHITE); else if(selectedCandidate==2)tft.drawBitmap(10, 30, elephant_logo,64,64,ILI9341_WHITE); else if(selectedCandidate==3)tft.drawBitmap(10, 30, bird_logo,64,64,ILI9341_WHITE); tft.setCursor(10, 110); tft.setTextColor(ILI9341_WHITE, ILI9341_BLACK); tft.println(candidates[selectedCandidate]); tft.setCursor(10, 140); tft.setTextColor(ILI9341_WHITE, ILI9341_BLACK); tft.print("Total votes: "); tft.println(votes[selectedCandidate]); }</pre>	
--	--

Table: Source Code for the main program

4 Implementation

4.1 Description

The slave device enables the voter to select candidate and cast the vote upon authentication and the master device calculates and displays the total vote counts for the candidates. Both of these devices were at first implemented separately and then the serial communication between the two devices was established.

Slave Device:

Major three components of the slave device included the Fingerprint Sensor for authentication, TFT Display for showing necessary instructions and the candidates, and the Rotary Encoder for smooth selection of candidates and casting the vote. We used an Arduino Mega as the microcontroller for the slave device.

The fingerprint sensor was interfaced with the arduino Mega with the following connections:

Sensor Pin		Connected Pin
VCC	→	Battery 5V
TX	→	Arduino RX1
RX	→	Arduino TX1
GND	→	Arduino GND

Figure: Pin Connection for fingerprint Sensor

Initially the enrollment of the voters were done using the Fingerprint Library modules. Next, authentication was also performed using the same library.

For the display module we used a 2.8" TFT display with the following connection:

Sensor Pin		Connected Pin
VCC	→	3.3V
GND	→	GND
CS	→	A5
Reset	→	A4
DC	→	A3
SDI	→	11
SCK	→	13
LED	→	3.3V
SDO	→	12

Figure: Pin Connection for TFT Display

10K resistors were used between the connections to limit the current. The name of the candidate was shown along with the corresponding symbol.

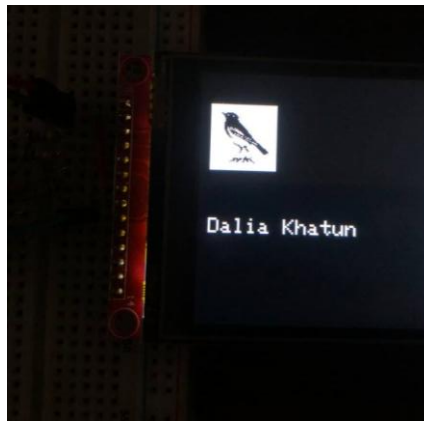


Figure: Display for the Slave Device

Using the rotary encoder, it was possible to develop a scalable system where the voter can easily rotate the encoder to select the desired candidate and cast the vote by using the built-in push button of the encoder.



Figure: Rotary Encoder

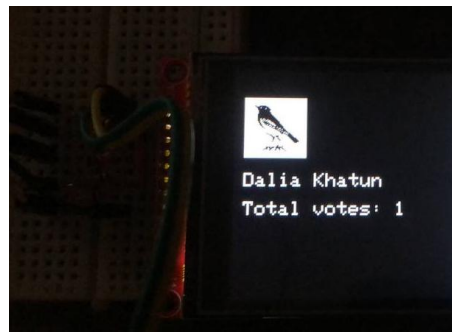
The rotary encoder has three pins apart from the VCC and GND pins. The three pins include CLK, DT and SW. The SW pin is responsible for the push button. Under normal condition, the SW pin is set to HIGH. When the button is pressed the SW pin outputs LOW.

The CLK and DT pins are responsible for capturing the rotation of the encoder. When rotated, the CLK pin changes its output state. For the direction of rotation, the following conditions are implied:

- If CLK changes & $DT \neq CLK$, Clockwise movement.
- If CLK changes & $DT = CLK$, Counter-Clockwise movement.

Master Device:

The master device has one major component, the display. By receiving the selected candidate's ID from the slave device via serial communication, the device maintains the total vote counts for each candidate. The display connection with the master device is similar to the slave device connection. But in this case, the total vote count for the candidates is also displayed.



We used UART communication (Serial communication) to transmit voting data from the voting Arduino (Slave) to the Master Arduino. It transfers data one bit at a time over a single data line without requiring a separate clock signal. It operates asynchronously meaning both devices must be configured to use the same baud rate (9600 bits per second). This method is very efficient and reliable for data transferring between microcontrollers. By implementing this communication system in our project, we ensured that all the votes are securely and reliably transmitted and handled at a central location.

UART operates by transmitting data as a series of bits, including a start bit, data bits, an optional parity bit, and stop bit(s).



Figure: UART frame structure

For UART communication, we connected the TX3, RX3 pins of our slave arduino to the defined RX and TX pins respectively of our master arduino and shorted the grounds. Whenever an authenticated voter casted their vote, the candidate id would be sent via UART communication protocol from the slave to master device. The master device then incremented the vote count of the corresponding candidate by 1.

4.2 Results and Analysis

Our project is based on two parts:

- ✓ Slave Device
- ✓ Master Device

Slave Device

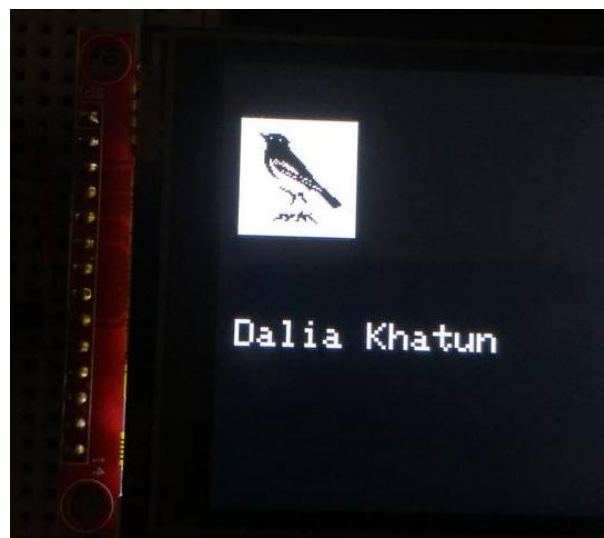
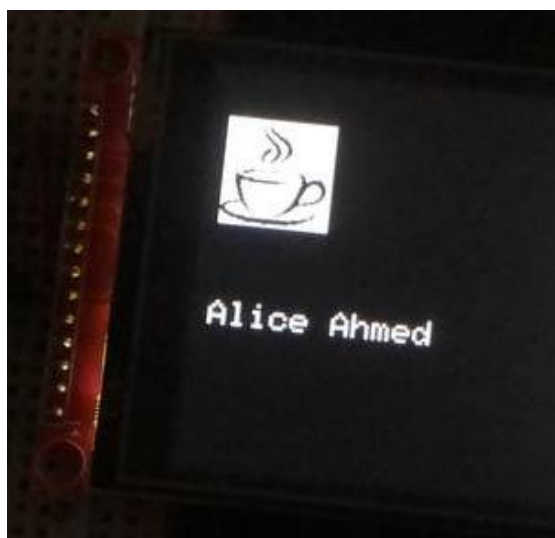
Fingerprint Sensor: In our fingerprint sensor, each voter's fingerprint is previously registered through proper procedure. So, when a verified voter will try to vote, his fingerprint will be scanned and matched against the saved records to confirm his identity. He will be able to cast a vote only if his fingerprint matches with the database. In our TFT display, his eligibility will be shown.

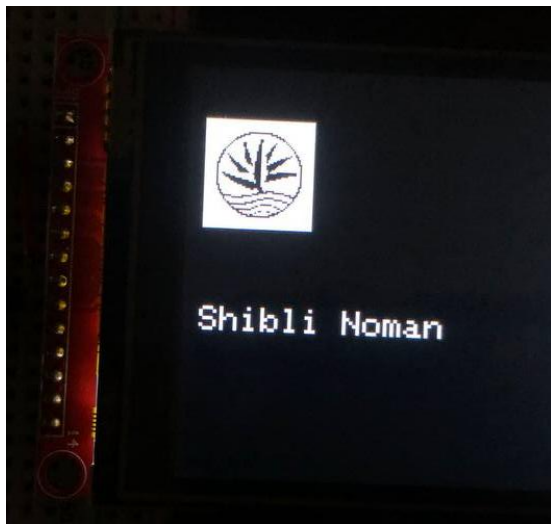


Rotary Encoder:

This component is used for navigating through the available options (candidate list) on the voting system. The verified user will scroll through all the choices shown on the TFT display by rotating the rotary knob and select a candidate by pressing the knob.

TFT Display:





From the above candidate's list, the voter will choose his favorite candidate. When the voter will select their favorite candidate from the list, their vote will be registered and the selected candidate's vote count will be incremented by one. It will ensure that each vote is accurately counted for the corresponding candidate.

The steps taken by the voter is shown on the Arduino Serial Monitor:

```
You can vote!  
  
Dalia Khatun  
Chayan Chowdhury  
Chayan Chowdhury is selected!  
  
You can vote!  
  
Dalia Khatun  
Chayan Chowdhury  
Shibli Noman  
Shibli Noman is selected!  
  
Already Voted!!!!  
  
You can vote!  
  
Dalia Khatun  
Chayan Chowdhury  
Shibli Noman  
Shibli Noman is selected!
```

Figure: Serial Monitor for slave device

A voter cannot cast a vote second time if he has already voted. A warning message will be displayed on the TFT screen, notifying the voter that they have already cast their vote.

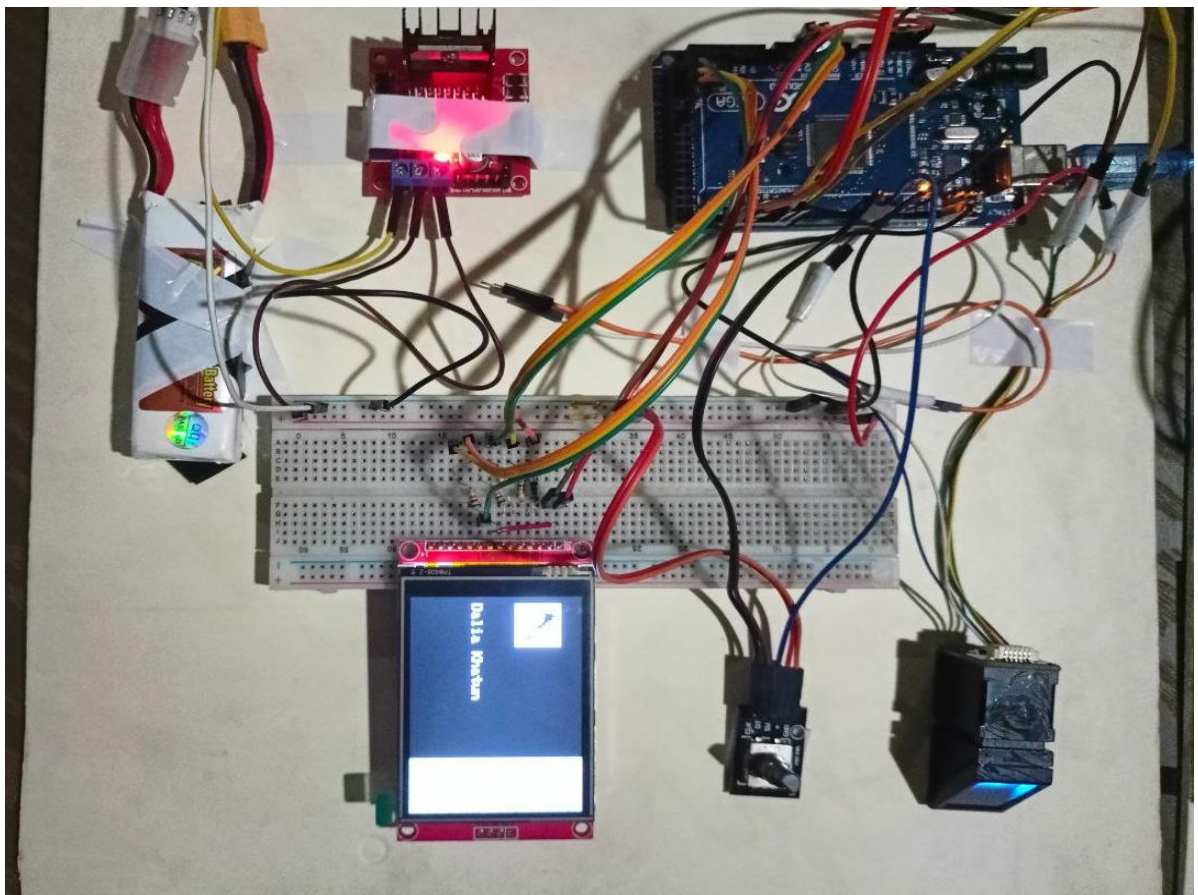
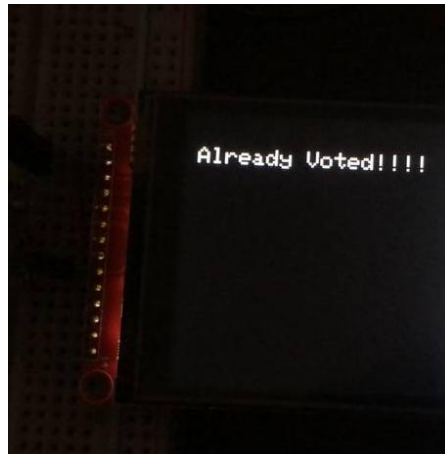


Figure: Complete Slave Device

Master Device

The master device will continuously monitor the vote count for each candidate, keeping track of all vote increments. The TFT display connected to the master Arduino will update

in real-time, showing the current vote count for each candidate.

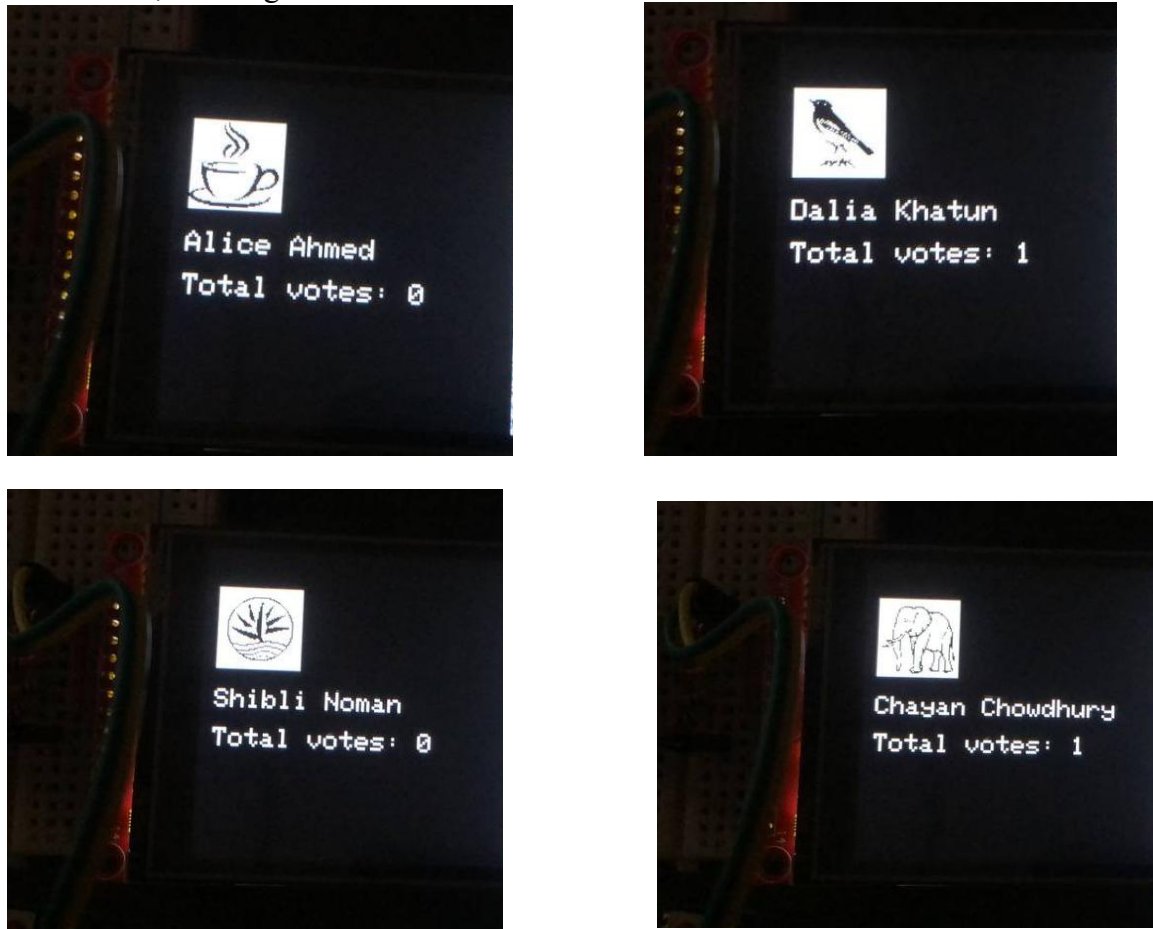


Figure: TFT display for master device

```
Vote for Chayan Chowdhury
Total Votes:
Alice Ahmed: 0 votes
Shibli Noman: 0 votes
Chayan Chowdhury: 1 votes
Dalia Khatun: 0 votes

Vote for Dalia Khatun
Total Votes:
Alice Ahmed: 0 votes
Shibli Noman: 0 votes
Chayan Chowdhury: 1 votes
Dalia Khatun: 1 votes
```

Figure: Serial Monitor for the master Device

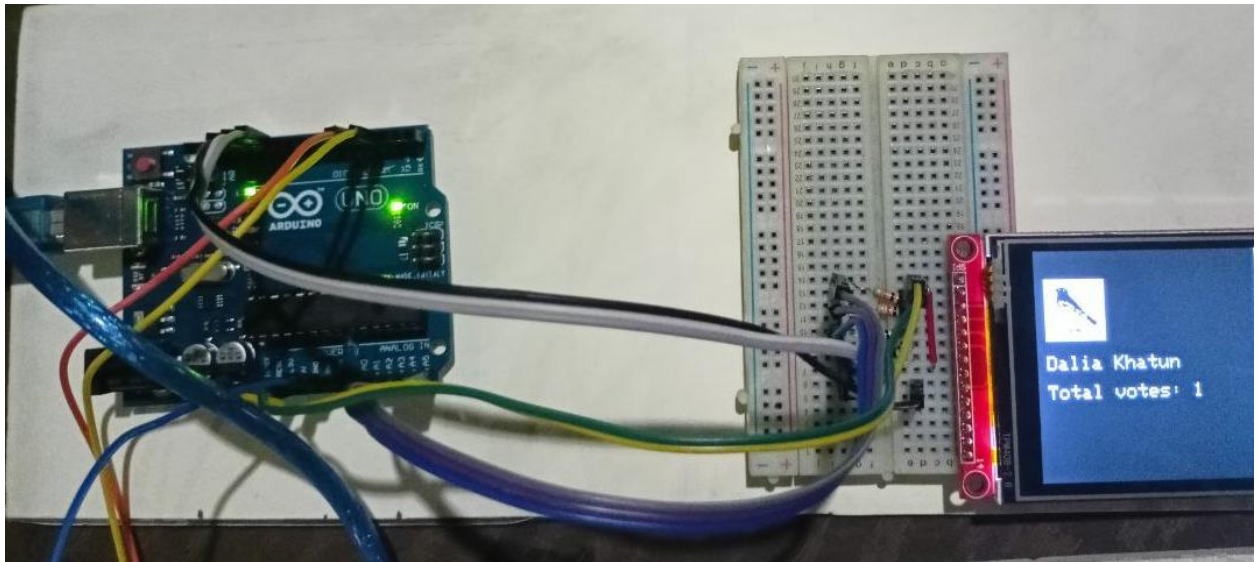


Figure: Master Device

Our Final Device

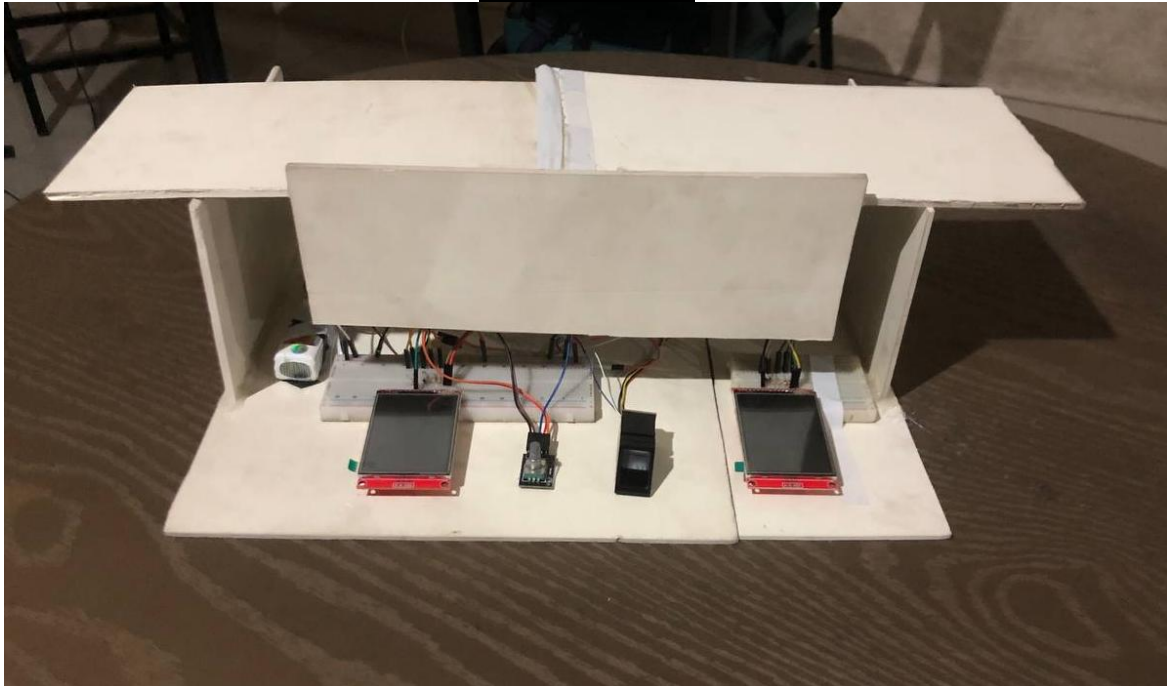


Figure: Slave Device (on the left), Master Device (on the right)

5 Design Analysis and Evaluation

5.1 Novelty

Our project introduces several new features that make it more advanced than traditional voting systems and basic electronic voting machines (EVMs). One of the main improvements is the use of a fingerprint sensor to check each voter's identity. This adds a strong layer of security, making sure that only eligible people can vote and preventing impersonation or multiple votes by the same person. Another unique part of our design is the use of a rotary encoder instead of fixed buttons to select candidates. This allows voters to scroll through a list of candidates, which makes it easier to support more candidates without changing the hardware layout. It also improves the user experience by making the selection process smoother and more flexible.

In addition, our system displays helpful instructions and live vote counts using TFT screens, giving both the voter and the officials a clear and real-time view of the process. We used UART serial communication to quickly and reliably send vote data from the voting device to the counting device. Lastly, we separated the system into two parts—a slave device for voting and a master device for counting—which makes the design more organized, easier to troubleshoot, and better for future upgrades. Altogether, these features make our project more secure, efficient, and user-friendly than traditional voting machines.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to Public Health and Safety

In our project, we made sure there are no risks to health or safety. Since everything was done using software and basic electronics, there was no use of dangerous materials or harmful equipment. We used low-voltage components like Arduino boards and fingerprint sensors, which are safe to handle. Also, by using fingerprint authentication, we made the voting process more secure and fair for everyone.

5.2.2 Considerations to Environment

We kept the environment in mind while designing the system. The entire project was done using reusable electronic parts, and there was no paper used, unlike traditional voting systems. This helps reduce waste. Also, our design uses low-power components, which means it consumes less electricity, making it energy-efficient and environmentally friendly.

5.2.3 Considerations to Cultural and Societal Needs

Our voting machine is designed to be simple and easy for everyone to use, no matter their background. The use of a rotary encoder and touchscreen makes the system accessible, even

for people who are not familiar with digital devices. By making voting easier, safer, and more transparent, our system supports democratic practices and can be useful in schools, communities, or local elections.

5.3 Limitations of Tools (PO(e))

While working on this project, we used tools like the Arduino IDE, Arduino UNO and Mega boards, and components such as TFT displays, fingerprint sensors, and rotary encoders. These tools were useful for basic design and testing, but they also had some limitations.

The Arduino boards, while easy to use, have limited processing power and memory. This can make it difficult to handle large fingerprint databases or more complex logic. The fingerprint sensor also had limited speed during verification, which might slow down the process when used by many voters. Additionally, serial communication (UART) worked well for small data transfers, but it may become slow or unstable if more devices or faster communication are needed.

We also noticed that the TFT display sometimes responded slowly or had limited resolution, which affects how much information we can show at once. Finally, the Arduino IDE does not support advanced debugging features, making it harder to find and fix complex issues in the code.

Despite these limitations, we were able to complete the project by working around these constraints and focusing on simple, efficient solutions.

5.4 Impact Assessment (PO(f))

Our project, the Electronic Voting Machine with Fingerprint Authentication, addresses several key societal and contextual concerns:

a. Public Health and Safety

By replacing manual voter authentication and paper ballots with biometric verification and digital voting, our system significantly reduces the chances of physical contact, tampering, and crowding—especially critical in scenarios like pandemics. The automated system also eliminates manual vote counting errors, preventing post-election disputes that can escalate into societal unrest.

b. Legal and Ethical Assurance

Voter impersonation is a major legal concern in elections. Our fingerprint-based authentication guarantees that only eligible and unique voters can cast their votes, thus upholding legal standards and electoral integrity. By reducing the human role in vote counting and result compilation, the system also minimizes manipulation and bias.

c. Societal and Cultural Needs

Voting is a deeply cultural and civic responsibility. This system respects that by ensuring equal access, transparency, and trust. It supports inclusivity regardless of language or literacy through its simple rotary encoder interface and clear visual display. In societies where election-related violence or fraud is common, this system can reduce societal tensions by providing verifiable and tamper-proof outcomes.

5.5 Sustainability Evaluation (PO(g))

Our project contributes meaningfully to both environmental and societal sustainability:

a. Environmental Sustainability

- **Paperless Voting:** The system eliminates paper ballots entirely, significantly reducing paper consumption and associated deforestation and waste.
- **Energy Efficiency:** All hardware components—microcontrollers, sensors, displays—are optimized for low power consumption. This reduces the carbon footprint of election activities and enables operation in remote or off-grid locations via small battery or solar power setups.

b. Societal Sustainability

Transparent Elections: By ensuring accuracy and preventing fraud, the system builds long-term public trust in the electoral process—crucial for democratic sustainability.

- **Scalability and Reusability:** Once developed, the system can be reused in multiple elections with minimal cost, making it financially and operationally sustainable for institutions or developing regions.
- **Conflict Reduction:** Transparent and verifiable voting processes minimize post-election conflicts and legal disputes, supporting a more peaceful and stable society.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

The team was first divided into sub groups where each group worked on separate parts of the project. Later, the entire team collaborated to implement the final circuit by ensuring proper integration.

Student ID	Contributions
2006055	Interfaced the rotary encoder module with the arduino, final integraion
2006056	Interfaced the fingerprint sensor, designed and implemented necessary algorithms and logic, final integration
2006057	Worked with display modules, designed and implemented necessary algorithms and logic, final integration
2006058	Interfaced the fingerprint sensor and implemented the serial communication between master and slave devices, final integration

6.2 Mode of Teamwork

Online Meetings: We utilized online platforms for regular meetings, allowing team members to connect regardless of location. This facilitated real-time discussions, brainstorming sessions, and quick decision-making, ensuring everyone stayed aligned on project goals.

- Offline Meetings: The team met in person and worked together to implement the various components and the final integration of the entire project.

6.3 Executive Summary

Our project provides a secure, transparent, low-cost and efficient solution to modernize the voting process. It replaces traditional ballot paper-based voting with a fully digital system that verifies voter identity using fingerprint biometrics and enables vote casting through a simple interface. The system consists of a slave device for user interaction and a master device for live vote counting via UART communication. This approach eliminates impersonation, reduces human error and paper waste, speeds up result processing. Environmentally friendly and socially impactful, the project ensures fair elections while promoting sustainability, trust, and accessibility in democratic practices.

7 Project Management and Cost Analysis (PO(k))

7.1 Bill of Materials

Component	Quantity	Unit Price (Tk)	Total Price (Tk)
Arduino UNO	1	650	650
Arduino Mega	1	1800	1800
TFT 2.8" Display	2	1140	2280
R307 Fingerprint Sensor	1	1950	1950
Rotary Encoder	1	110	110
Battery (Lipo)	1	1350	1350
Resistors	6-10	-	15
Breadboard	2	145	290
Jumper Wires	1 set	100	100
PVC Boards	2	110	220
		Total=	8765

7.2 Logbook of Project Implementation:

Date	Milestone achieved	Individual Role (Last two digits of Student ID are used)	Comments
01/05/25	Formation of team for project		The team defined its goal and started looking for relevant project ideas.
04/05/25	Selection of Project		The team expressed their project ideas to the course teachers. The project was then finalized, and feedbacks were provided to us.
18/05/25	Project Proposal presentation		A formal presentation on the project idea was given and necessary instructions were provided to us by the course teachers
01/06/25	Project plan		A final plan for the project was done by the team via online meeting on zoom
18/06/25	Division of tasks		Individual tasks were assigned to the members
26/06/25	Offline meeting before project update	55: Implemented the circuit and code for rotary encoder. 56: Implemented the code and logic for master and slave devices 57: Implemented the circuit and code for the display modules 58: Implemented the circuit and code for serial communication	Each member presented their work, that were done individually, to others and integrated the entire circuit
29/06/25	Project Update Presentation	56 and 58 prepared the presentation slide for the update	A project update presentation was given in front of the course teachers which showed the work we had done so far and how we

			proceed to go from there. Feedbacks were provided to us
24/07/25	Final project implementation	55 and 57 worked on the changes in the display modules that were suggested by the course teachers during the update. 56 and 58 worked on the fingerprint sensor 56 and 57 worked on the overall logic and code for the two devices All the members then worked to build the final version of the project	The team worked together for the final project integration
27/07/25	Final Project Demonstration/Presentation		The demonstration of our final project to our course instructors was given.

8 Future Work (PO(I))

- We can connect our project to the internet, allowing real-time vote data to be displayed on a website. Through the website, anyone can view the live vote count for each candidate, providing transparency and accessibility to the voting process.
- In addition to the fingerprint sensor, we can integrate other biometric options for voter verification, such as facial recognition or iris scanning. This will provide multiple methods of authentication, enhancing the security and reliability of the voting system.
- We can develop a mobile application for voters to check their voting status, view candidate information, and receive updates on the voting process.
- We can enhance accessibility by adding features for people with disabilities, such as voice-controlled voting or screen readers for visually impaired voters.

9 References

- “Arduino - Rotary Encoder” – <https://www.instructables.com/Arduino-Rotary-Encoder-Simple-Example-KY-040/>
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- “UART: A Hardware Communication Protocol Understanding Universal Asynchronous Receiver/Transmitter” - <https://www.analog.com/en/resources/analog-dialogue/articles/uart-a-hardware-communication-protocol.html>
- “Basics of UART communication” - <https://www.circuitbasics.com/basics-uart-communication/>
- “Interfacing fingerprint sensor with arduino” - <https://www.instructables.com/Interfacing-Fingerprint-Sensor-With-Arduino/>
- Arduino Serial documentation – “<https://docs.arduino.cc/language-reference/en/functions/communication/serial/>”